



رياضيات متقطعة
د. كاء شالعة

الرياضيات المتقطعة العددية

المحاضرة الرابعة

المرحلة الثالثة

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Lecture: Graphs and Trees in Discrete Mathematics

1. Introduction

graphs and trees are essential structures in discrete mathematics. They are used to model relationships between objects, such as computer networks, social networks, transportation systems, and data structures.

A graph consists of vertices (nodes) and edges (connections) that link these vertices.

2. Basic Definitions

Graph (G)

A graph $= G(V, E)$ consists of

a set of vertices (or nodes) $:V$ •

a set of edges, where each edge connects two vertices $:E$ •

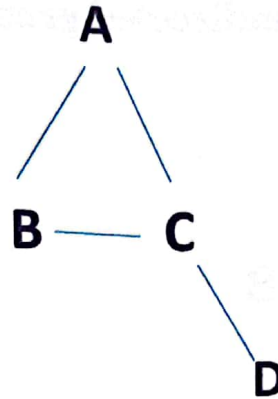
:Example

Let

and $\{A, B, C, D\} = V$

$\{\{C, D\}, \{B, D\}, \{A, C\}, \{A, B\}\} = E$

:We can represent this graph visually



3. Degree of a Vertex

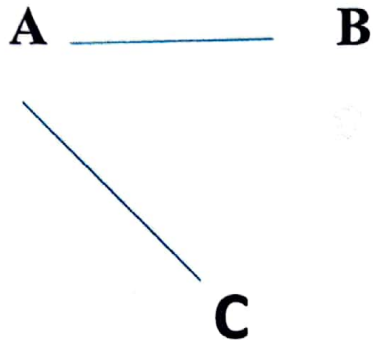
- Degree of a vertex = number of edges incident to it.
- In directed graphs:
 - In-degree = number of incoming edges.
 - Out-degree = number of outgoing edges.

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Example:

In the following undirected graph:



$$\deg(A) = 2$$

$$\deg(B) = 1$$

$$\deg(C) = 1$$

عدد الفروع لل (A ,B,C)

4. Paths and Cycles

- Path: A sequence of edges that connect a sequence of distinct vertices.
- Cycle: A path that starts and ends at the same vertex.

Example

In the graph below:

A - B - C - D - A

Path: $A \rightarrow B \rightarrow C \rightarrow D$

Cycle: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$

5. Trees

A tree is a connected graph with no cycles.

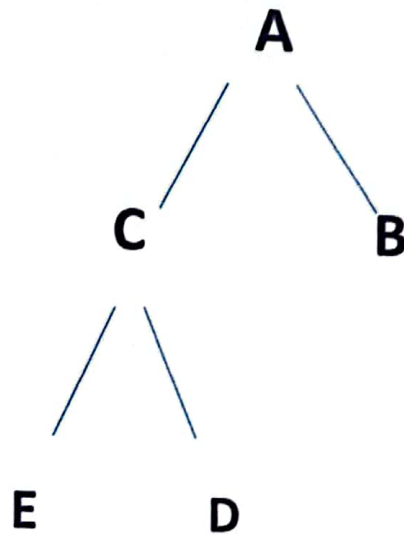
It's one of the most important structures in computer science.

Properties of Trees

1. A tree with n vertices has $n-1$ edges.
2. There is exactly one path between any two vertices.
3. Removing any edge disconnects the tree.

Example 1: Identify a Tree

Consider the following graph



Vertices = 5

Edges = 4

There are no cycles, and it is connected.

✓ This is a tree.

الرؤوس هس = 5

والحواف او الفروع = 4

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Example 2: Is it a Tree or Not

A - B - C - D - A

Vertices = 4

Edges = 4

There is a cycle (A → B → C → D → A)

✗ Not a tree.